



Girraween High School

Year 12 HSC Trial Examination

August 2014

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Circle the letter corresponding to the correct answer.

Question 1 (1 mark)

What is the correct value of $|-1| - |-1|$?

- A. 0
- B. -1
- C. 1
- D. 2

Question 2 (1 mark)

What is 0.01077 rounded to 3 significant figures?

- A. 0.0107
- B. 0.011
- C. 0.0108
- D. 0.01

Question 3 (1 mark)

Solve the equation $3x - 1 = \frac{x + 3}{2}$

- A. $x = \frac{4}{5}$
- B. $x = 1$
- C. $x = 2$
- D. $x = 7$

Question 4 (1 mark)

Which is the correct factorisation of $3x^2 + x - 2$?

- A. $(3x + 2)(x - 1)$
- B. $(3x - 2)(x + 1)$
- C. $(3x - 1)(x + 2)$
- D. $(3x + 1)(x - 2)$

Question 5 (1 mark)

What is the distance between the points $(6, 1)$ and $(3, -3)$?

- A. 5
- B. 25
- C. $\sqrt{7}$
- D. 7

Question 6 (1 mark)

When the denominator is rationalised, $\frac{1}{\sqrt{3} - \sqrt{2}} =$

- A. $\frac{\sqrt{3} - \sqrt{2}}{5}$
- B. $\sqrt{3} - \sqrt{2}$
- C. $\frac{\sqrt{3} + \sqrt{2}}{5}$
- D. $\sqrt{3} + \sqrt{2}$

Question 7 (1 mark)

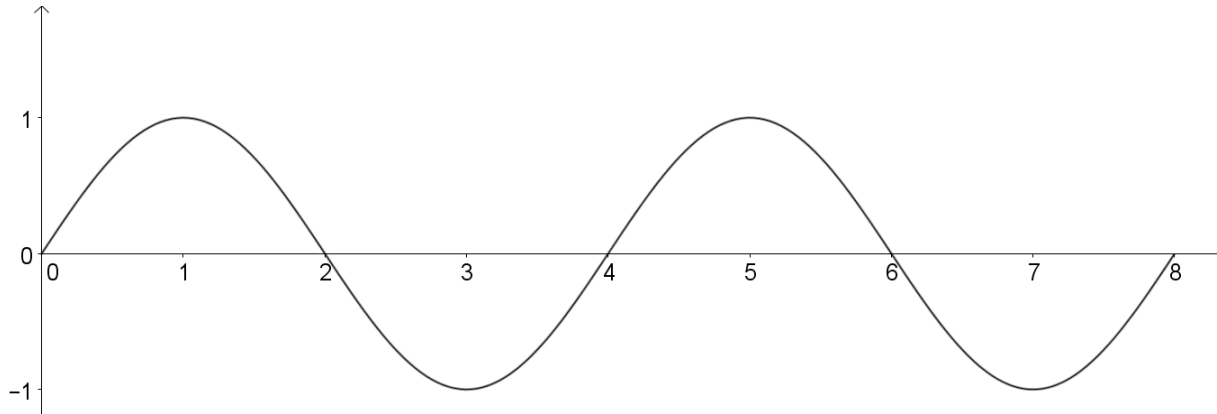
What is the domain of the function $f(x) = \sqrt{6 - 2x}$?

- A. All real x such that $x \leq 3$
- B. All real x such that $x < 3$
- C. All real x such that $x \geq 3$
- D. All real x such that $x > 3$

Question 8 on the next page

Question 8 (1 mark)

Below is a graph of $y = \sin \frac{\pi x}{2}$. Which of the following integrals has the greatest value?



- A. $\int_0^1 \sin \frac{\pi x}{2} dx$
- B. $\int_0^2 \sin \frac{\pi x}{2} dx$
- C. $\int_0^4 \sin \frac{\pi x}{2} dx$
- D. $\int_0^7 \sin \frac{\pi x}{2} dx$

Question 9 (1 mark)

When x is replaced with $x + 1$ in the equation $y = x^2$, the graph is moved:

- A. One unit to the right
- B. One unit to the left
- C. One unit higher
- D. One unit lower

Question 10 (1 mark)

Which of the following is the correct function value at the minimum turning point of:

$$f(x) = (x - 2012)(x - 2013)(x - 2014)$$

- A. -1
- B. 0
- C. $\frac{1}{2}(2012)(2013)(2014)$
- D. $\frac{-2\sqrt{3}}{9}$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Write your answers on the paper provided.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

(a) Evaluate $\sqrt{\pi^2 + 5}$ to two decimal places. [1]

(b) Convert $\frac{3\pi}{5}$ radians to degrees. [1]

(c) Find the exact value of $\sin \frac{2\pi}{3}$ [1]

(d) Simplify $\frac{x}{x^2 - 4} + \frac{2}{x - 2}$ [2]

(e) Find the values of x for which $|x - 3| \geq 3$ [2]

(f) Differentiate with respect to x

i. $y = x^2 \ln 3x$ [2]

ii. $y = \frac{\sin 4x}{x^3}$ [2]

(g) Find $\int 1 + \sec^2 x \, dx$ [2]

(h) Find the limiting sum of the geometric series [2]

$$\frac{3}{4} + \frac{3}{16} + \frac{3}{64} + \dots$$

Question 12 (15 marks)

- (a) A function $f(x)$ passes through the point $(2, 10)$. Given that [3]

$$f'(x) = 3x^2 - 3x + 5$$

find the value of $f(1)$.

- (b) In a certain arithmetic series, the first term is 13 and the sixth term is -7 .

i. Find the common difference. [1]

ii. Find the value of the third term. [2]

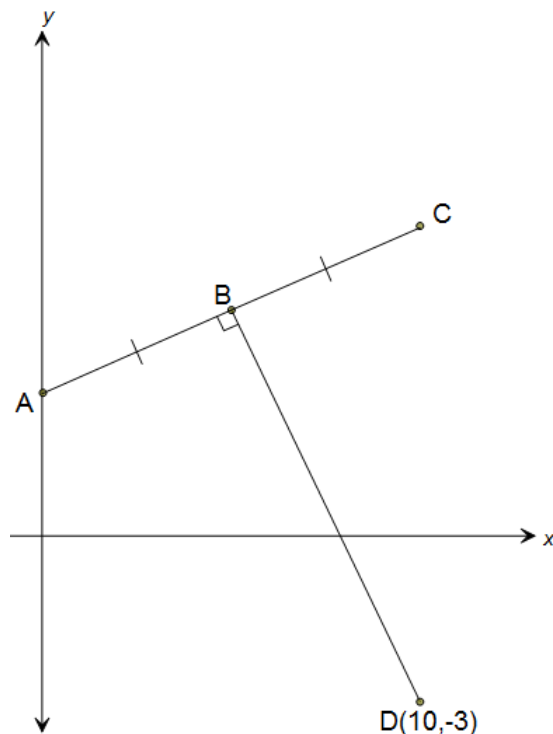
- (c) Consider the parabola $y + 2 = 8(x - 1)^2$. Find:

i. the coordinates of the vertex, [1]

ii. the coordinates of the focus, [2]

iii. the equation of the directrix. [1]

- (d) The diagram shows points A , B and C lying on the line $2y = x + 4$. The point A lies on the y -axis and $AB = BC$. The line from $D(10, -3)$ to B is perpendicular to AC .



i. Find the coordinates of A . [1]

ii. Find the equation of the line BD . [2]

iii. Find the coordinates of C . [2]

Question 13 (15 marks)

(a) Let α and β be the roots of $2x^2 - 4x - 2 = 0$.

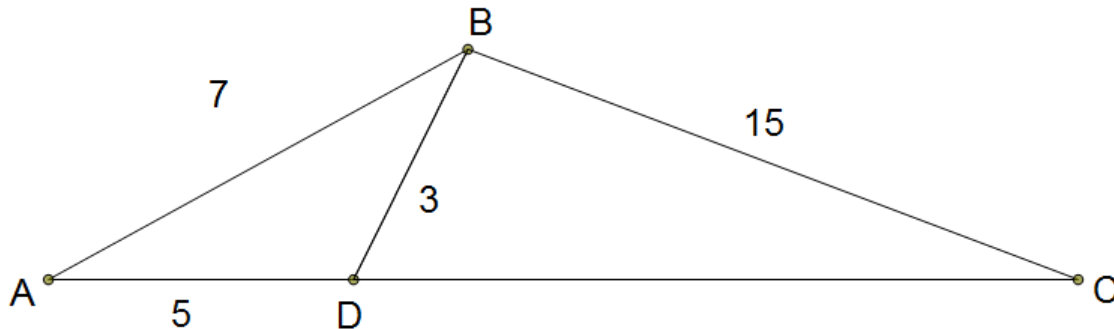
i. State the value of $\alpha\beta$

[1]

ii. Find $\frac{5}{\alpha} + \frac{5}{\beta}$

[2]

(b) In the diagram below, triangle ABC has dimensions $AB = 7\text{cm}$ and $BC = 15\text{cm}$. The point D lies on AC such that $AD = 5\text{cm}$ and $BD = 3\text{cm}$.



i. Use the cosine rule to show that $\angle ADB = 120^\circ$.

[2]

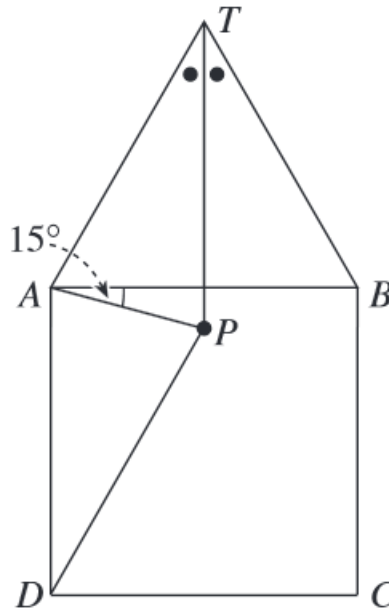
ii. Show that $\angle BCD = 10^\circ$ (rounded to the nearest degree).

[2]

iii. Find the length of DC , correct to the nearest cm.

[2]

(c) In the diagram below, $ABCD$ is a square and ABT is an equilateral triangle. The line TP bisects $\angle ATB$, and $\angle PAB = 15^\circ$.



i. Copy the diagram into your writing booklet and explain why $\angle PAT = 75^\circ$.

[1]

ii. Prove that $\triangle TAP \equiv \triangle DAP$.

[3]

iii. Prove that $\triangle DAP$ is isosceles.

[2]

Question 14 (15 marks)

- (a) Consider the function $f(x) = x^4 - 4x^3$.
- i. Show that $f'(x) = 4x^2(x - 3)$ [1]
 - ii. Find the coordinates of the stationary points of the curve $f(x)$, and determine their nature. [3]
 - iii. Sketch the graph of the curve $f(x)$, showing the stationary points. [1]
 - iv. Find the values of x for which the graph of $f(x)$ is concave down. [2]

- (b) Jenny borrows \$500 000 to buy a house. An interest rate of 9% p.a. compounded monthly is charged on the outstanding balance. The loan is to be repaid in equal monthly instalments of \$ M over a 25 year period (300 months). Let A_n be the amount owing after n months.
- i. Show the amount owing after 3 months is: [2]

$$A_3 = 500000 \times 1.0075^3 - M(1 + 1.0075 + 1.0075^2)$$

- ii. Find the required monthly repayment. [2]
 - iii. How much interest does Jenny pay over the 25 years? [1]
- (c) A bag contains six discs. Two of the discs have the number 0 on them and the other four discs have the number 1 on them. Three discs are drawn at random without replacement.
- i. Find the probability that all of the three discs drawn have the number 1 on them. [2]
 - ii. Find the probability that the product of the numbers on the three discs drawn is 0. [1]

The exam continues on the next page

Question 15 (15 marks)

(a) Suppose $y = \sqrt{3^x + x}$:

- i. Complete the table below, giving the values of y to 3 decimal places. [2]

x	0	0.25	0.5	0.75	1
y	1	1.251			2

- ii. Use the trapezoidal rule with all the values of y from the table above to find an approximation for the value of [3]

$$\int_0^1 \sqrt{3^x + x} \, dx$$

(b) The mass (kg) of a decaying substance at time t years is given by $M = M_0 e^{-kt}$, where M_0 is its initial mass and k is a positive constant.

- i. Show that M satisfies the differential equation $\frac{dM}{dt} = -kM$. [1]

- ii. Show that the half-life of the substance is given by $\frac{\ln 2}{k}$ years. [2]

- iii. A second substance is also decaying simultaneously with the first substance but its rate of decay is twice as fast. Find the mass of the second radioactive substance N in terms of M_0 and k , given that its initial mass is half of the first substance. [3]

(c) The velocity of a particle moving along the x axis at time t is given by

$$v = te^{2t}$$

- i. Find the acceleration of the particle. [1]
- ii. Given that when $t = 0$ the particle is at $x = 0$, find x in terms of t . [3]

The exam continues on the next page

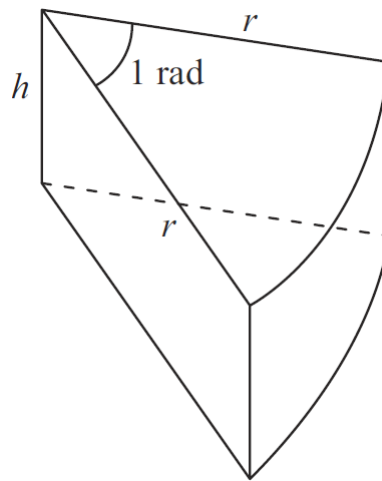
Question 16 (15 marks)

(a) i. Given $2\log_3 x - \log_3(x - 2) = 2$ show that x satisfies $x^2 - 9x + 18 = 0$. [3]

ii. Hence, or otherwise, solve the equation: [1]

$$2\log_3 x - \log_3(x - 2) = 2$$

(b) The diagram below shows a closed box used by a shop for packing pieces of cake. The box is a right prism of height h cm. The cross section is a sector of a circle. The sector has radius r cm and angle 1 radian. The volume of the box is 300 cm^3 .



i. Show that the surface area of the box, $S \text{ cm}^2$, is given by: [3]

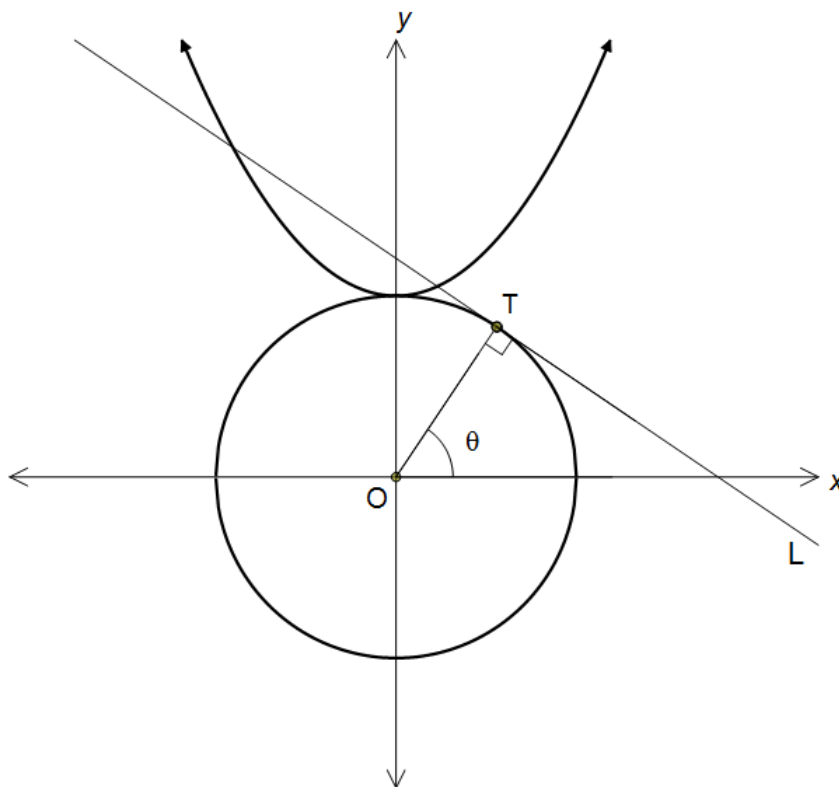
$$S = r^2 + \frac{1800}{r}$$

ii. Find the value of r that minimises the the value of S . Give your answer correct to one decimal place. [3]

The exam continues on the next page

- (c) The diagram below shows the graph of $y = x^2 + 1$ and a point T on the unit circle $x^2 + y^2 = 1$ at angle θ from the positive x -axis, where $0 \leq \theta \leq 2\pi$.

The tangent line L to the circle at T is perpendicular to OT .



- i. Show that the equation of the tangent line L is given by [2]

$$x \cos \theta + y \sin \theta = 1$$

- ii. Show that if L intersects $y = x^2 + 1$ twice then $\sin \theta$ satisfies the following inequalities [2]

$$-\frac{1}{5} < \sin \theta < 0 \text{ or } 0 < \sin \theta < 1$$

- iii. Find the values of θ to the nearest minute such that L intersects $y = x^2 + 1$ twice. [1]

End of exam

Year 12 HSC TRIAL 2014

Multiple Choice Solutions:

1. A 2. C 3. B 4. B 5. A

6. D 7. A 8. B 9. B 10. D

Q1

$$|-1| - |-1| = 1 - 1 = 0 \quad \therefore \textcircled{A}$$

Q2

$$= 0.0108 \quad \therefore \textcircled{C}$$

Q3

$$3x - 1 = \frac{x+3}{2}$$

$$6x - 2 = x + 3$$

$$6x - x = 2 + 3$$

$$5x = 5$$

$$x = 1 \quad \therefore \textcircled{B}$$

Q4

$$3x^2 + x - 2$$

$$3x \quad -2$$

$$x \quad 1$$

$$(3x - 2)(x + 1) \quad \therefore \textcircled{B}$$

Q5

$$D = \sqrt{(6-3)^2 + (1+3)^2}$$

$$= \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\therefore \textcircled{A}$$

Q6

$$\frac{1}{\sqrt{3}-\sqrt{2}} \times \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}+\sqrt{2}}$$

$$= \frac{\sqrt{3}+\sqrt{2}}{3-2} = \sqrt{3}+\sqrt{2} \quad \therefore \textcircled{D}$$

Q7

$$6 - 2n \geq 0$$

$$6 \geq 2n$$

$$2n \leq 6$$

$$n \leq 3 \quad \therefore \textcircled{A}$$

Q8

$$\int_0^4 y \, dx = 0$$

$$0 < \int_0^1 y \, dx = \int_0^7 y \, dx$$

$$\int_0^2 y \, dx = 2 \int_0^1 y \, dx$$

$$\therefore \textcircled{B}$$

Q9

x replaced with $x+1$ moves the graph 1 unit to the left.

$$\therefore \textcircled{B}$$

Q10

The value and the nature of turning points does not change under translations of the graph.

So replacing x with $x+2013$ in $f(x)$ we get:

$$f(x) = (x+1)x(x-1)$$

$$f(x) = x(x+1)(x-1)$$

$$f(x) = x(x^2-1)$$

$$f(x) = x^3 - x$$

$$f'(x) = 3x^2 - 1$$

$$f'(x) = 0 \text{ when } 3x^2 - 1 = 0$$

$$x^2 = \frac{1}{3} \quad x = \pm \frac{1}{\sqrt{3}}$$

$$f''(x) = 6x \quad \therefore f''\left(\frac{1}{\sqrt{3}}\right) = 6 \times \frac{1}{\sqrt{3}} > 0$$

So $x = \frac{1}{\sqrt{3}}$ gives the minimum

$$f\left(\frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}} \left(\frac{1}{3} - 1\right) = \frac{1}{\sqrt{3}} \times -\frac{2}{3}$$

$$= \frac{-2}{3\sqrt{3}} = \frac{-2\sqrt{3}}{9}$$

$\therefore$ (D)

Q11

(a) 3.86 (2 dp).

(b) $\frac{3\pi}{5} = \frac{3\pi}{5} \times \frac{180}{\pi} = 108^\circ$

(c) $\sin \frac{2\pi}{3} = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$

(d) $\frac{x}{x^2-4} + \frac{2}{x-2}$

$$= \frac{x}{x^2-4} + \frac{2(x+2)}{x^2-4}$$

$$= \frac{x+2x+4}{x^2-4} = \frac{3x+4}{x^2-4}$$

(e) $|x-3| \geq 3$

$$x-3 \geq 3 \quad \& \quad 3-x \geq 3$$

$$x \geq 6 \quad \& \quad 3-3 \geq x$$

$$x \leq 0$$

(f)

(i) $y = x^2 / \ln 3x$

$$u = x^2 \quad v = \ln 3x$$

$$u' = 2x \quad v' = \frac{1}{x}$$

$$y' = 2x / \ln 3x + x$$

Q11

(f)

$$(ii) y = \frac{\sin 4x}{x^3}$$

$$u = \sin 4x \quad v = x^3$$

$$u' = 4\cos 4x \quad v' = 3x^2$$

$$y' = \frac{4x^3 \cos 4x - 3x^2 \sin 4x}{x^6}$$

$$y' = \frac{4x \cos 4x - 3 \sin 4x}{x^4}$$

$$(g) \int 1 + \sec^2 x \, dx$$

$$= x + \tan x + C$$

$$(h) a = \frac{3}{4} \quad r = \frac{1}{4}$$

$$S_{\infty} = \frac{a}{1-r} = \frac{\frac{3}{4}}{1-\frac{1}{4}} = \frac{\frac{3}{4}}{\frac{3}{4}} = 1$$

Q12

$$(a) f'(x) = 3x^2 - 3x + 5$$

$$f(x) = x^3 - \frac{3}{2}x^2 + 5x + C$$

$$f(2) = 10$$

$$\therefore 10 = 2^3 - \frac{3}{2} \times 2^2 + 5 \times 2 + C$$

$$10 = 8 - 6 + 10 + C$$

$$C = 6 - 8 = -2$$

$$\therefore f(x) = x^3 - \frac{3}{2}x^2 + 5x - 2$$

$$\therefore f(1) = 1 - \frac{3}{2} + 5 - 2$$

$$f(1) = \frac{5}{2}$$

(b)

$$i. a = 13$$

$$T_6 = a + (6-1)d$$

$$-7 = 13 + 5d$$

$$5d = -20, \text{ so } d = -4$$

$$ii. T_3 = 13 + 2 \times -4 = 13 - 8 = 5$$

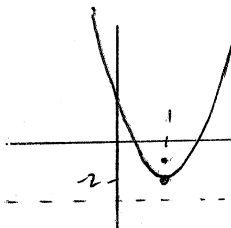
(c)

$$(i) \frac{1}{8}(y+2) = (x-1)^2$$

$$\text{So } V = (1, -2)$$

$$(ii) 4a = \frac{1}{8} \text{ so } a = \frac{1}{32}$$

$$\therefore F = (1, -2 + \frac{1}{32}) = (1, -\frac{63}{32})$$



$$(iii) D: y = -2\frac{1}{32}$$

$$(d) y = \frac{x}{2} + 2$$

(i)

$$\text{So } A = (0, 2)$$

$$(ii) m \text{ of } BD = -2 \text{ as } -2 \times \frac{1}{2} = -1$$

$$\text{So } m = -2 \text{ \& point } = (10, -3)$$

$$y + 3 = -2(x - 10)$$

$$y = -2x + 20 - 3$$

$$y = -2x + 17$$

Q12

(d)

(iii) Finding intersection of line AC

& BD:

$$2y = x + 4$$

$$y = -2x + 17$$

$$\text{So } 2(-2x + 17) = x + 4$$

$$-4x + 34 = x + 4$$

$$30 = 5x$$

$$x = 6$$

$$\therefore y = -2 \times 6 + 17 = 5$$

$$\therefore B = (6, 5)$$

Since B is the mid point of AC

$$6 = \frac{x+0}{2} \quad \& \quad 5 = \frac{2+y}{2}$$

$$\therefore x = 12 \quad 2+y = 10$$

$$y = 8$$

$$\therefore C = (12, 8)$$

Q13

(a)

$$(i) \alpha\beta = \frac{c}{a} = \frac{-2}{2} = -1$$

$$(ii) \frac{5}{\alpha} + \frac{5}{\beta} = \frac{5\beta + 5\alpha}{\alpha\beta}$$

$$= \frac{5(\alpha + \beta)}{\alpha\beta}$$

$$= \frac{5 \times \frac{-(-4)}{2}}{-1}$$

$$= \frac{10}{-1} = -10$$

(b)

$$(i) \cos \angle ADB = \frac{5^2 + 3^2 - 7^2}{2 \times 5 \times 3}$$

$$\cos \angle ADB = -\frac{1}{2}$$

$$\therefore \angle ADB = 180 - 60 = 120^\circ$$

(ii) $\angle BDC = 60$ ($\angle$ in a straight line)

Using sine rule in $\triangle BDC$:

$$\frac{\sin \angle BCD}{3} = \frac{\sin 60}{15}$$

$$\sin \angle BCD = \frac{\sqrt{3}}{2} \times \frac{1}{5} = \frac{\sqrt{3}}{10}$$

$$\angle BCD = \sin^{-1}\left(\frac{\sqrt{3}}{10}\right) = 10^\circ \text{ (nearest deg.)}$$

$$(iii) \frac{DC}{\sin \angle DBC} = \frac{15}{\sin 60}$$

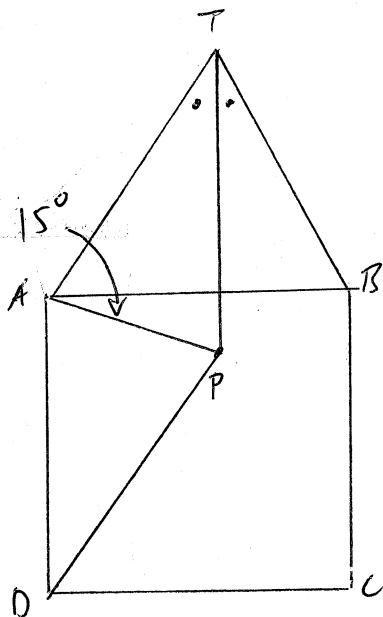
$$\angle DBC = 180 - 60 - \angle BCD \text{ ($\angle$ sum of a $\triangle$)}$$

$$DC = \frac{15 \sin \angle DBC}{\sin 60} = 16 \text{ cm (nearest cm)}$$

Q13

(c)

(i)



$\angle TAB = 60^\circ$ ($\angle$ in an equilateral Δ)

$$\therefore \angle PAT = 60 + 15 = 75^\circ$$

(ii)

$AD = AB$ (Sides of a square)

$AB = AT$ (Sides of an equilateral Δ)

$$\therefore AD = AT$$

$$\angle DAP = 90 - 15 = 75^\circ \text{ ($\angle$ in a square)}$$

$$\therefore \angle DAP = \angle PAT = 75^\circ$$

AP is common.

$$\therefore \Delta TAP \equiv \Delta DAP \text{ (SAS)}$$

(iii) $\angle ATP = 30^\circ$ (TP bisects $\angle ATB$)

$$\therefore \angle TPA = 180 - 30 - 75 = 75^\circ$$

($\angle$ sum of Δ)

$\therefore AT = TP$ (Equal sides opposite equal $\angle$'s in isosceles ΔATP)

But since $\Delta APD \equiv \Delta ATP$,

ΔAPD must also be

Isosceles.

Q14

(a)

$$(i) f(x) = x^4 - 4x^3 = x^3(x-4)$$

$$f'(x) = 4x^3 - 12x^2 \\ = 4x^2(x-3)$$

$$(ii) f'(x) = 0 \text{ when } 4x^2(x-3) = 0$$

$$\text{So } x = 0 \text{ or } x = 3$$

$$f(0) = 0 \text{ \& } f(3) = -27$$

So stat pts are $(0,0)$ & $(3,-27)$

$$f''(x) = 12x^2 - 24x$$

$$f''(x) = 12x(x-2)$$

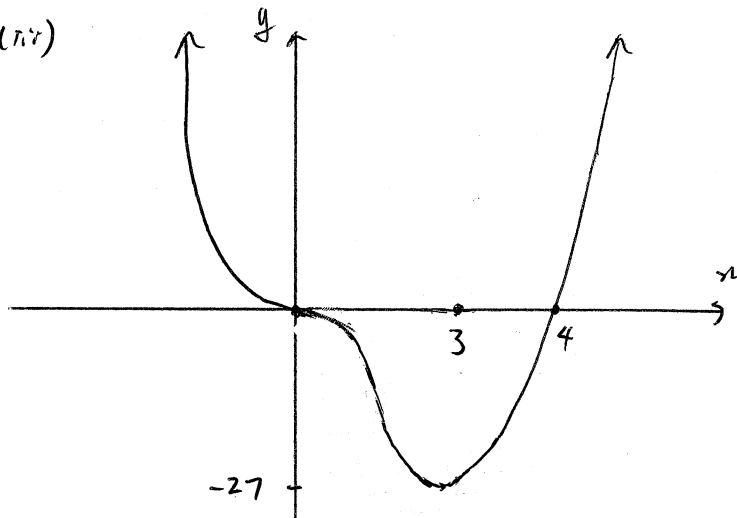
x	-1	0	1
$f'(x)$	-16	0	-8

$\therefore (0,0)$ is a horizontal point of inflection.

$$f''(3) = 36 > 0 \therefore (3, -27)$$

a local minimum.

(iii)



(iv) concave down when $f''(x) < 0$

Q14

(b)

$$(i) R = \frac{9}{12} \div 100 = 0.0075.$$

$$A_1 = 500000(1.0075) - M$$

$$A_2 = A_1(1.0075) - M$$

$$A_2 = 500000(1.0075)^2 - M(1.0075) - M$$

$$A_3 = A_2(1.0075) - M$$

$$= 500000(1.0075)^3 - M(1.0075)^2 \\ - M(1.0075) - M$$

$$= 500000(1.0075)^3 - M(1 + 1.0075 + 1.0075^2)$$

(ii)

$$A_n = 500000(1.0075)^n - M(1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{n-1})$$

$$A_{300} = 500000(1.0075)^{300} - M \left(\frac{1.0075^{300} - 1}{1.0075 - 1} \right)$$

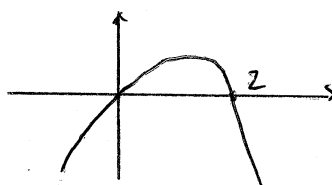
$$A_{300} = 0 \text{ when:}$$

$$M = 500000(1.0075)^{300} \times \frac{0.0075}{1.0075^{300} - 1}$$

$$M = \$4195.98$$

$$(iii) I = 4195.98 \times 300 - 500000 \\ = \$758794.55$$

$$\text{So } 12x(x-2) < 0$$



$$\text{So } 0 < x < 2$$

Q14

(c)

$$(i) p = \frac{4}{6} \times \frac{3}{5} \times \frac{2}{4} = \frac{1}{5}$$

$$(ii) p(\text{at least one } 0) = 1 - p(\text{all } 1's) \\ = 1 - \frac{1}{5} = \frac{4}{5}$$

Q15

(a)

(i)

x	0	0.25	0.5	0.75	1
y	1	1.251	1.494	1.741	2

$$(ii) h = 0.25$$

$$A \approx \frac{0.25}{2} \left[1 + 2 \times 1.251 + 2 \times 1.494 + 2 \times 1.741 + 2 \right]$$

$$A \approx 1.4965$$

(b)

$$(i) M = M_0 e^{-kt}$$

$$\frac{dM}{dt} = -k M_0 e^{-kt} \\ = -k M$$

$$(ii) \frac{1}{2} M_0 = M_0 e^{-kt}$$

$$\frac{1}{2} = e^{-kt}$$

$$-kt = \ln \frac{1}{2}$$

$$-kt = \ln 1 - \ln 2$$

$$kt = \ln 2 \quad \therefore t = \frac{\ln 2}{k}$$

(iii)

$$\frac{dN}{dt} = -2k M_0 e^{-kt}$$

$$\therefore N = -2k M_0 \int e^{-kt} dt$$

$$N = 2 M_0 e^{-kt} + C$$

$$\text{When } t=0, N = \frac{1}{2} M_0$$

$$\therefore \frac{1}{2} M_0 = 2 M_0 + C$$

$$\therefore C = -\frac{3}{2} M_0$$

$$\therefore N = 2 M_0 e^{-kt} - \frac{3}{2} M_0$$

(c)

$$(i) v = t e^{2t}$$

$$u = t \quad v = e^{2t}$$

$$u' = 1 \quad v' = 2e^{2t}$$

$$a = e^{2t} + 2te^{2t}$$

$$(ii) \int a dt = v$$

$$\text{So } \int e^{2t} + 2te^{2t} dt = te^{2t}$$

$$\text{So } \int e^{2t} dt + 2 \int te^{2t} dt = te^{2t}$$

$$\text{So } \frac{1}{2} e^{2t} + 2 \int v dt = te^{2t}$$

Q15

(c)

$$(ii) \quad 2 \int v dt = te^{2t} - \frac{1}{2}e^{2t}$$

$$2x = te^{2t} - \frac{1}{2}e^{2t}$$

$$x = \frac{1}{2}te^{2t} - \frac{1}{4}e^{2t} + C$$

$$t=0 \quad x=0$$

$$0 = 0 - \frac{1}{4} + C \quad \therefore C = \frac{1}{4}$$

$$\therefore x = \frac{1}{2}te^{2t} - \frac{1}{4}e^{2t} + \frac{1}{4}$$

Q16

$$(a) \quad 2\log_3 x - \log_3 (x-2) = 2$$

$$(i) \quad \log_3 (x^2) - \log_3 (x-2) = 2$$

$$\log_3 \left(\frac{x^2}{x-2} \right) = 2$$

$$\frac{x^2}{x-2} = 3^2$$

$$x^2 = 9(x-2)$$

$$x^2 = 9x - 18$$

$$x^2 - 9x + 18 = 0$$

$$(ii) \quad x^2 - 9x + 18 = 0$$

$$x \quad -3$$

$$x \quad -6$$

$$(x-3)(x-6) = 0$$

$$x=3, x=6$$

(b)

(i)

$$\begin{aligned} \text{Area of cross section} &= \frac{1}{2} \times l \times r^2 \\ &= \frac{1}{2} r^2 \end{aligned}$$

$$\therefore V = \frac{1}{2} h r^2$$

$$\therefore 300 = \frac{1}{2} h r^2$$

$$\therefore h = \frac{600}{r^2}$$

$$S = 2\left(\frac{1}{2}r^2\right) + 2(hr) + (rh)$$

$$S = r^2 + 2hr + rh$$

$$S = r^2 + 3hr$$

$$S = r^2 + \frac{1800}{r}$$

$$(ii) \quad S = r^2 + 1800r^{-1}$$

$$S' = 2r - 1800r^{-2}$$

$$S' = 2r - \frac{1800}{r^2}$$

$$S' = 0 \quad \text{when} \quad 2r = \frac{1800}{r^2}$$

$$2r^3 = 1800$$

$$r^3 = 900$$

$$r = \sqrt[3]{900}$$

$$S'' = 2 + 3600r^{-3} = 2 + \frac{3600}{r^3}$$

$$S'' = 2 + \frac{3600}{900} > 0 \quad \therefore r = \sqrt[3]{900}$$

$$r = 9.7 \text{ cm (1 dp)}$$

gives the minimum

Q16

(c)

$$(i) T = (\cos \theta, \sin \theta)$$

$$m_{OT} = \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\therefore m \text{ of } L = -\frac{\cos \theta}{\sin \theta}$$

$\therefore$ Equation of L is given by

$$y - \sin \theta = -\frac{\cos \theta}{\sin \theta} (x - \cos \theta)$$

$$y \sin \theta - \sin^2 \theta = -\cos \theta (x - \cos \theta)$$

$$y \sin \theta - \sin^2 \theta = -x \cos \theta + \cos^2 \theta$$

$$y \sin \theta + x \cos \theta = \sin^2 \theta + \cos^2 \theta$$

$$x \cos \theta + y \sin \theta = 1$$

$$\text{when } \theta = 0, \quad x + y \sin 0 = 1$$

$$x = 1$$

$$\text{when } \theta = 180, \quad x \cos 180 + y \sin 180 = 1$$

$$-x + 0 = 1$$

$$x = -1$$

$$\therefore L \text{ is given by } x \cos \theta + y \sin \theta = 1.$$

(ii)

$$x \cos \theta + y \sin \theta = 1$$

$$y = x^2 + 1$$

For intersections:

$$x \cos \theta + (x^2 + 1) \sin \theta = 1.$$

$$(\cos \theta)x + (\sin \theta)x^2 + \sin \theta - 1 = 0.$$

$$(\sin \theta)x^2 + (\cos \theta)x + (\sin \theta - 1) = 0.$$

$$x = \frac{-\cos \theta \pm \sqrt{\cos^2 \theta - 4(\sin \theta)(\sin \theta - 1)}}{2 \sin \theta}$$

for $\sin \theta \neq 0$.

$$\text{So } \Delta = \cos^2 \theta - 4(\sin \theta)(\sin \theta - 1)$$

$$= 1 - \sin^2 \theta - 4\sin^2 \theta + 4\sin \theta$$

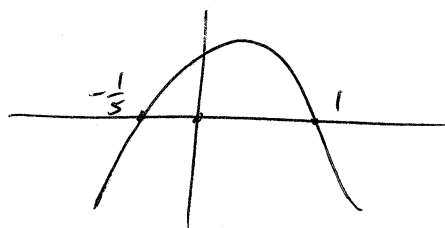
$$= -5\sin^2 \theta + 4\sin \theta + 1$$

So 2 intersections when

$$-5\sin^2 \theta + 4\sin \theta + 1 > 0.$$

$$\begin{array}{rcl} -5\sin \theta & & -1 \\ \sin \theta & & -1 \end{array}$$

$$(-5\sin \theta - 1)(\sin \theta - 1) > 0$$



$$\text{So } -\frac{1}{5} < \sin \theta < 1 \text{ but } \sin \theta \neq 0$$

$$\text{So } -\frac{1}{5} < \sin \theta < 0 \text{ \& } 0 < \sin \theta < 1.$$

Q16/

(c)

(iii) Solving $\sin \theta = -\frac{1}{5}$

$$\theta = 180 + \sin^{-1}\left(\frac{1}{5}\right) \text{ and}$$

$$\theta = 360 - \sin^{-1}\left(\frac{1}{5}\right)$$

$$\theta = 191^{\circ} 32' \text{ and } 348^{\circ} 27'$$

$$\theta = 191^{\circ} 32' \text{ and } -11^{\circ} 32'$$

$$\therefore -11^{\circ} 32' < \theta < 191^{\circ} 32'$$

$$\text{and } \theta \neq 0^{\circ} \text{ and } \theta \neq 180^{\circ}$$